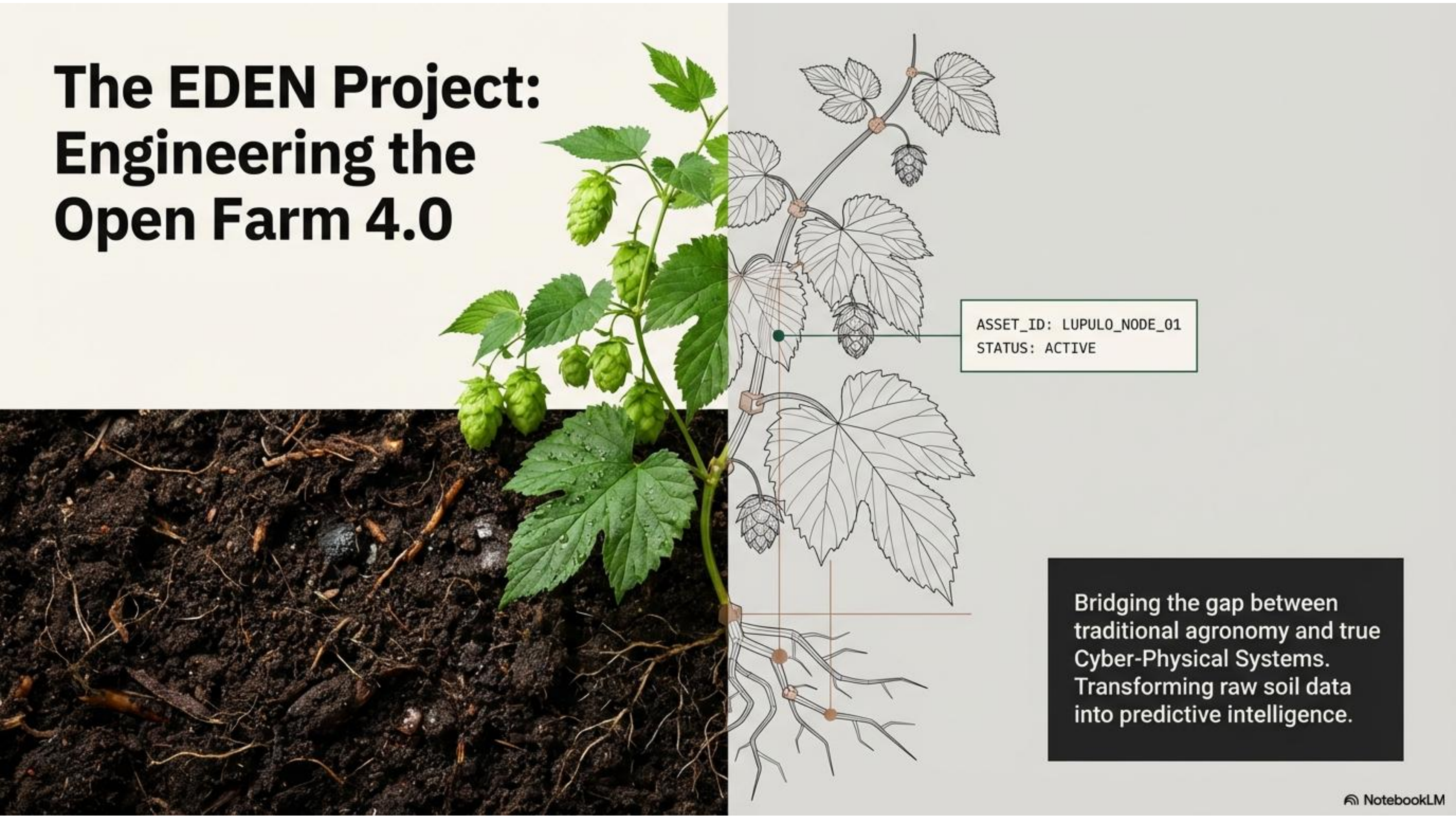


The EDEN Project: Engineering the Open Farm 4.0



ASSET_ID: LUPULO_NODE_01
STATUS: ACTIVE

Bridging the gap between
traditional agronomy and true
Cyber-Physical Systems.
Transforming raw soil data
into predictive intelligence.

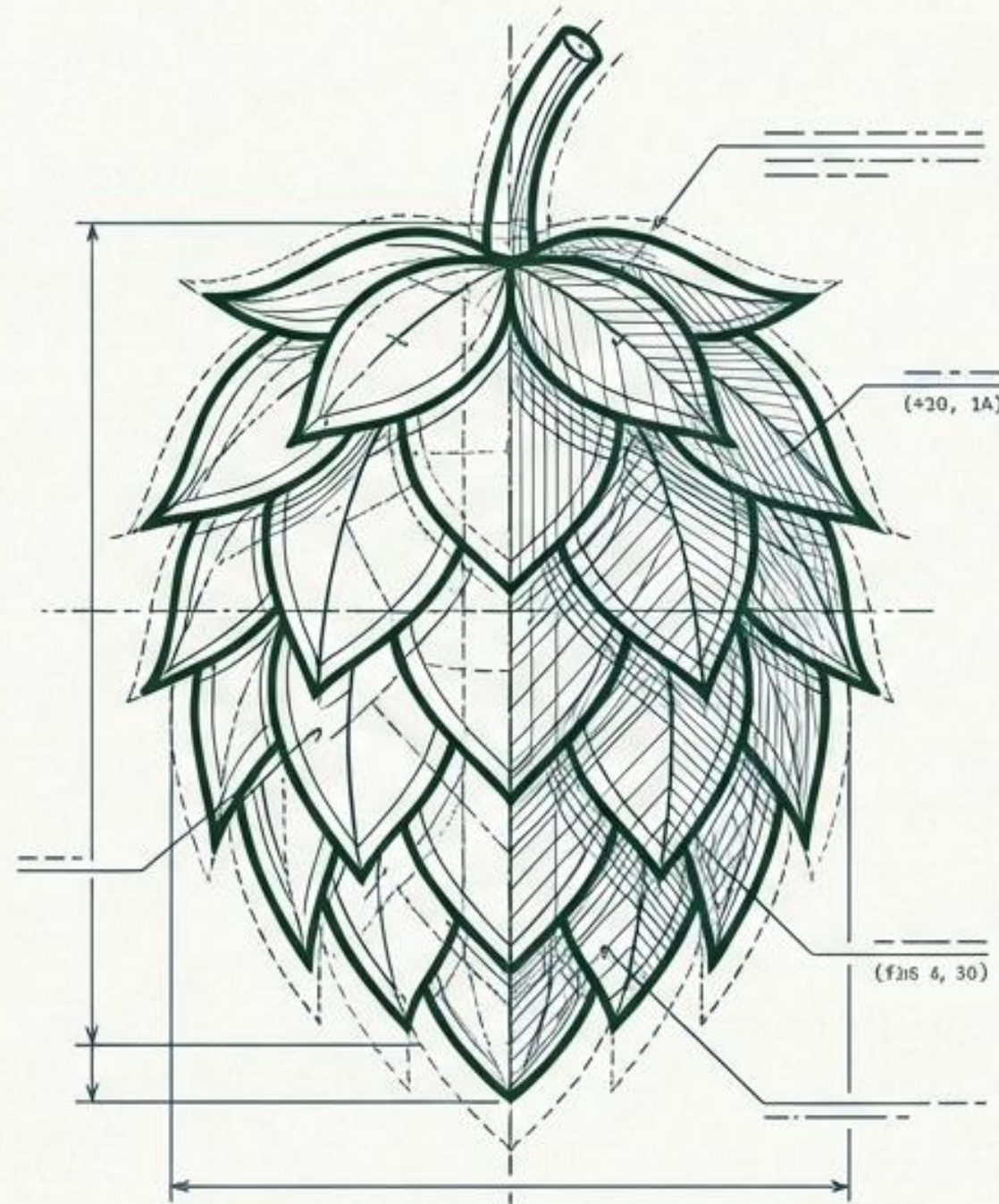
Project Éden Hops: From Theory to Soil

Mission

Develop an IoT sensor station integrated with Machine Learning to optimize precision hop farming.

Methodology

Applied, quantitative-qualitative experimental research. Moving from raw field sensing to predictive yield modeling.



Scope

Continuous collection and processing of environmental and soil variables to predict negative scenarios via supervised learning.

Vision

Empower small-to-medium farmers to produce export-quality hops domestically, proving that high-yield precision agriculture is accessible.

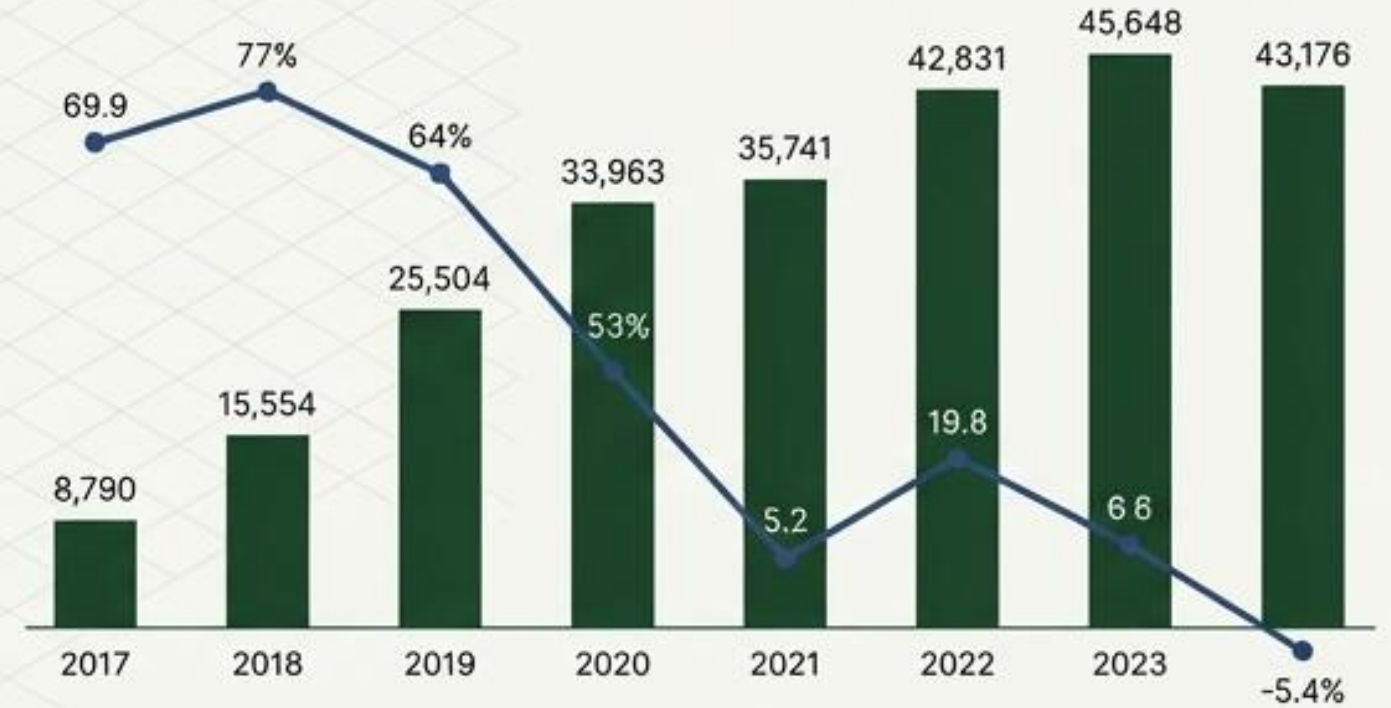
The Micro Context: The Brazilian Beer & Hop Paradox



The Market Boom

Metric: 43,176 registered beers in Brazil as of 2024

Insight: A massive, thriving national industry demanding high-quality agricultural inputs.



The Agricultural Bottleneck

Metric: ~100% Import Dependency

Insight: Hops—the crucial ingredient for aroma and bitterness—are almost entirely imported due to their requirement for highly precise micro-climates.

The Opportunity: Localizing hop cultivation empowers small producers and strengthens the national economy. Technology is the only way to replicate and guarantee these strict micro-climate conditions.

The System's Gravity: Physical vs. Digital Barriers

The evolution of AgTech faces a war on two fronts. **Hardware design** must **overcome mutating biology**. **Data architectures** must combat human failures and instinctive distrust.



The Physical Friction

- Material degradation in corrosive and humid soil.
- Physical incompatibility with accelerated plant growth.
- Severe fluctuations in power supply in the field.



The Digital Friction

- Fragmentation into data silos from isolated manufacturers.
- Low adoption: Farmer rejection of complex interfaces.
- Failures in security protocols and information governance.

Open Farming Suffers from Disconnected Intelligence

The Problem



Fragmented Data Silos:
Sensors exist, but they do not speak
a universal language.

The EDEN Anchor



The EDEN Proving Ground: Hop
(*Humulus lupulus*)
Cultivation.

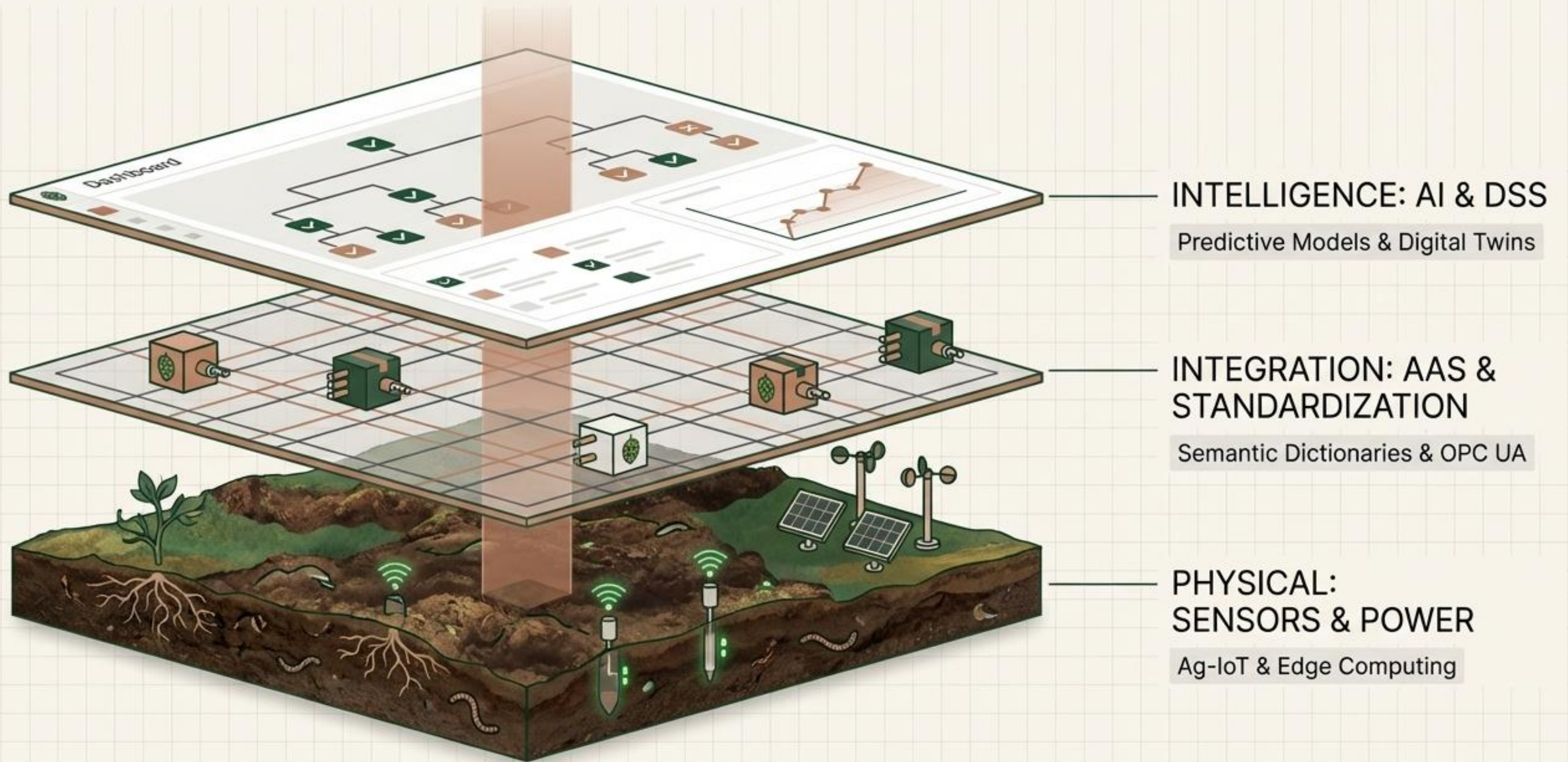
Highly sensitive to
temperature, humidity,
and soil pH. The perfect
candidate for predictive
transformation.

TENP SENSITIVITY: HIGH
HUNIDITY RANGE: 65-75%
pH OPTIMUM: 6.0-7.5
CULTIVAR: CASCADE

The Paradigm Shift to Agriculture 4.0

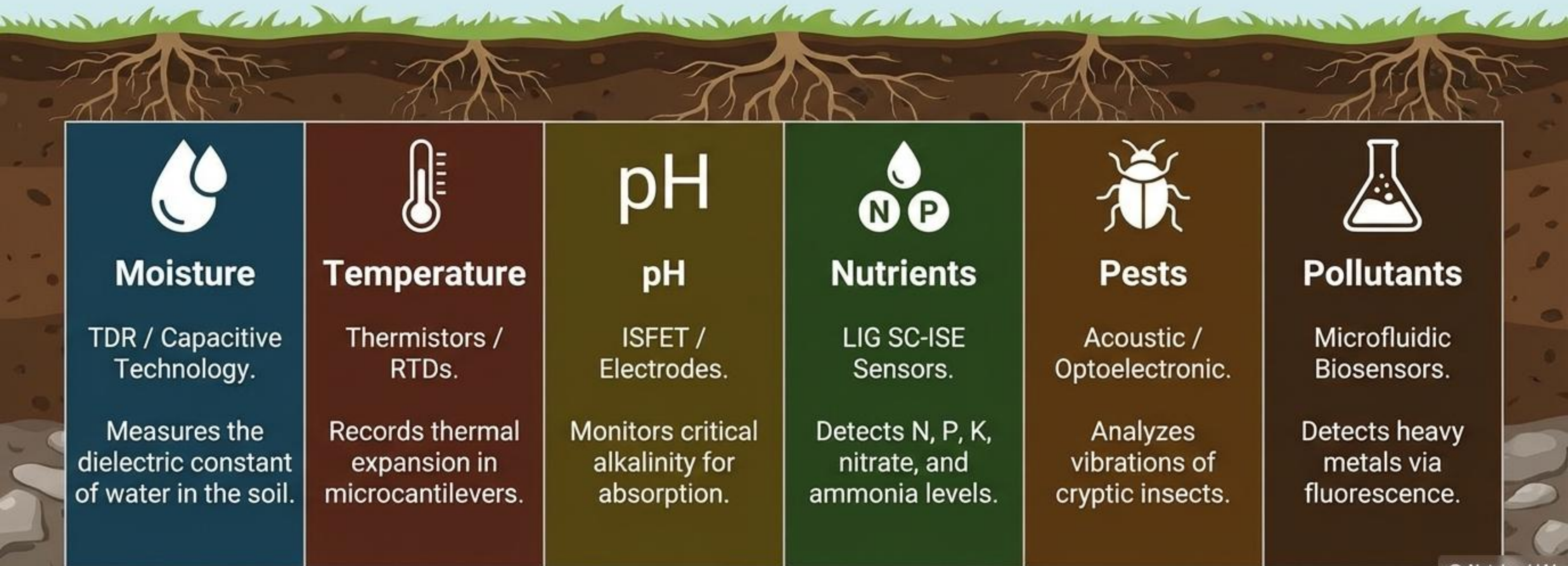
	Traditional	Ag-Tech 3.0	EDEN (Ag 4.0)
Reaction Time	Reactive (Wait for visible crop stress)	Real-Time (Alerts when thresholds are breached)	Predictive (Anticipates stress before it occurs)
Data Structure	Human observation/paper	Proprietary vendor silos	Standardized via Asset Administration Shells (AAS)
Decision Maker	Human guesswork	Human analysis of dashboard charts	AI-driven Decision Support System (DSS)

The EDEN Architecture Stack

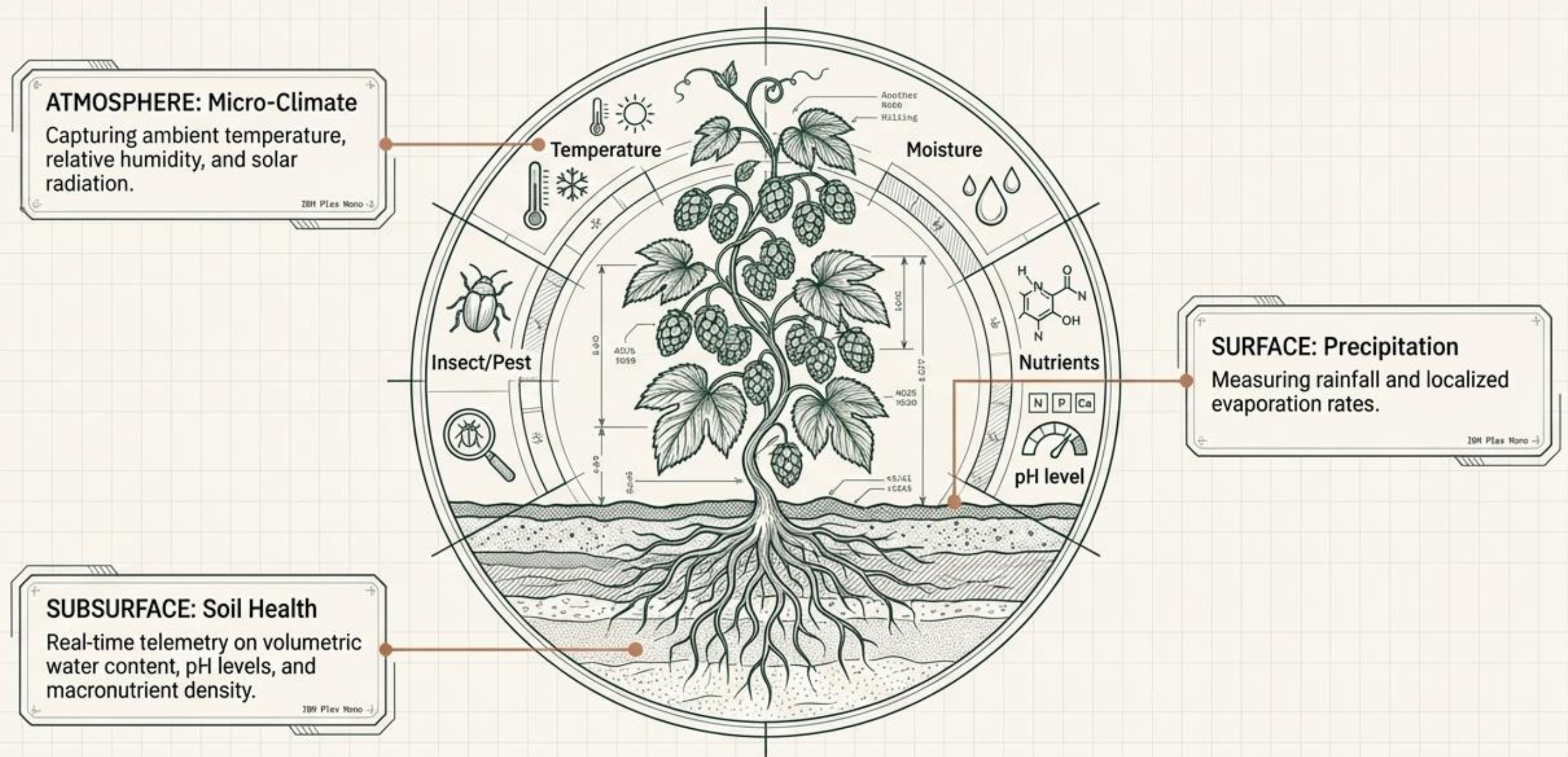


The Periodic Table of Soil Health

Intelligence begins at the root. The agricultural lithosphere is no longer a black box; it is perfectly quantifiable. Matrices of embedded nanosensors measure parameters with previously impossible spatiotemporal resolution.

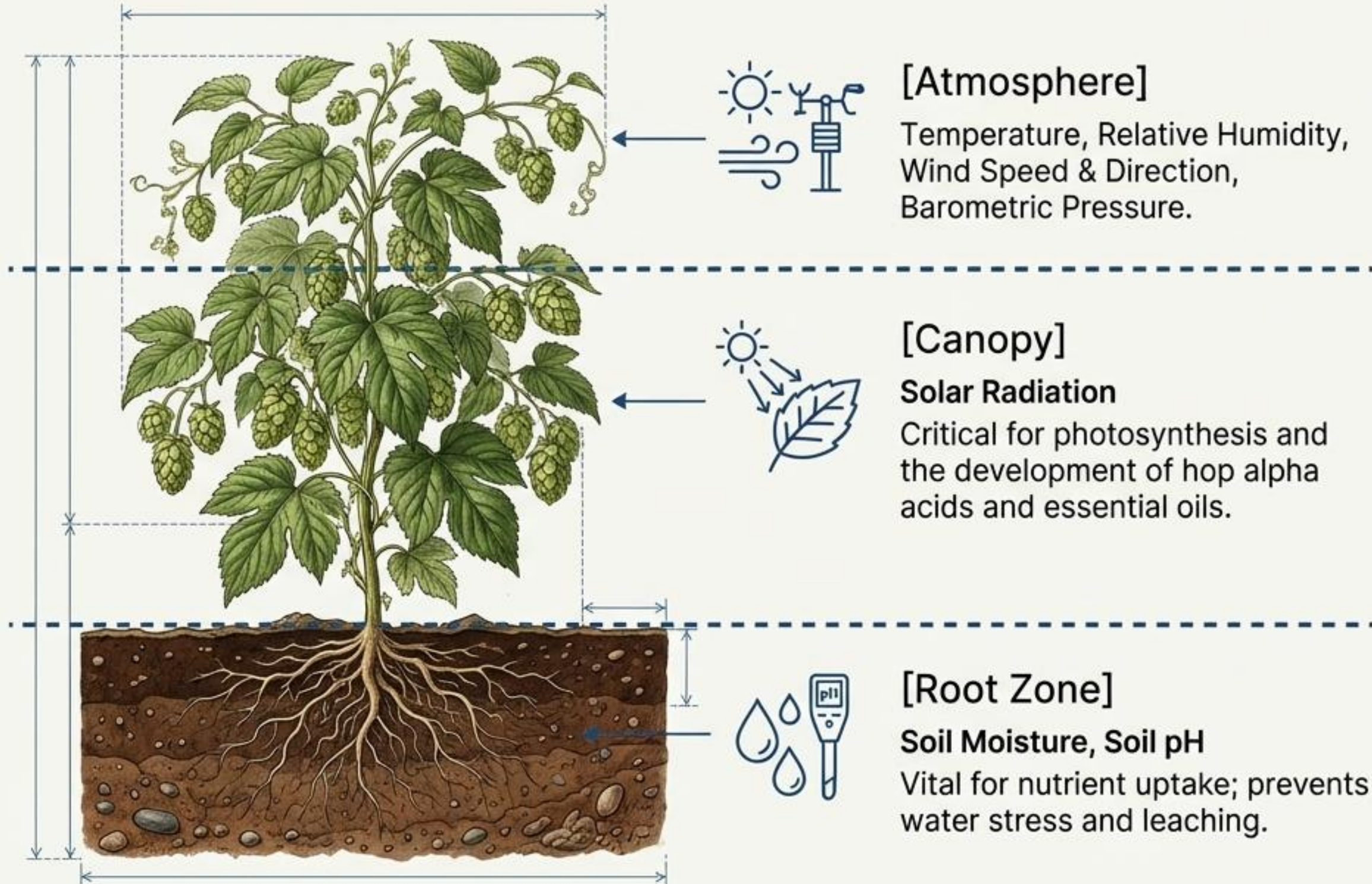


Step 1: Capturing the Physical Reality of the Hop Vine



Raw data is useless without context. It must first survive the constraints of the open field.

High-Resolution Sensing: The Micro-Environment



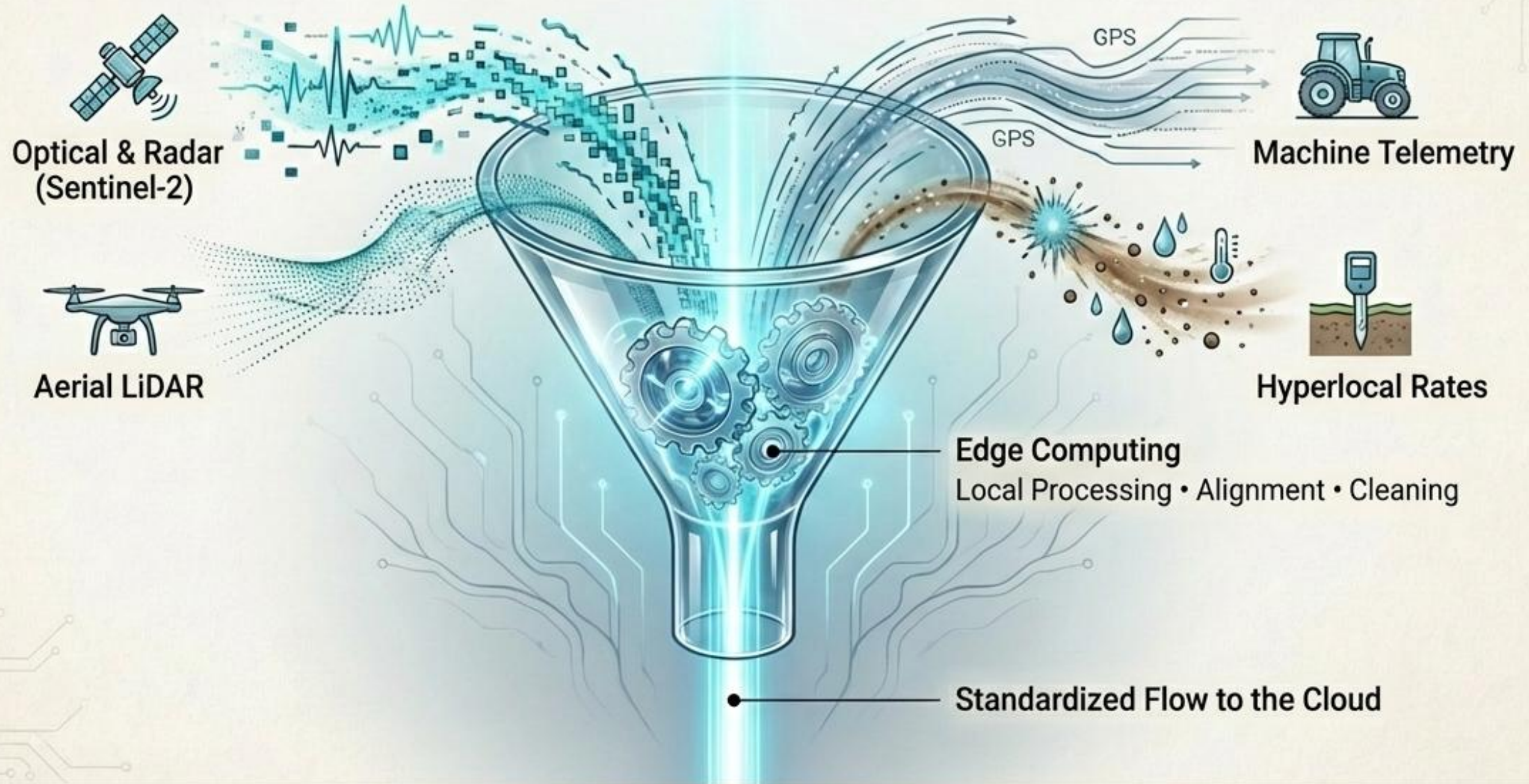
The Paradigm Shift:

Traditional farming treats an entire field as a single, uniform entity.

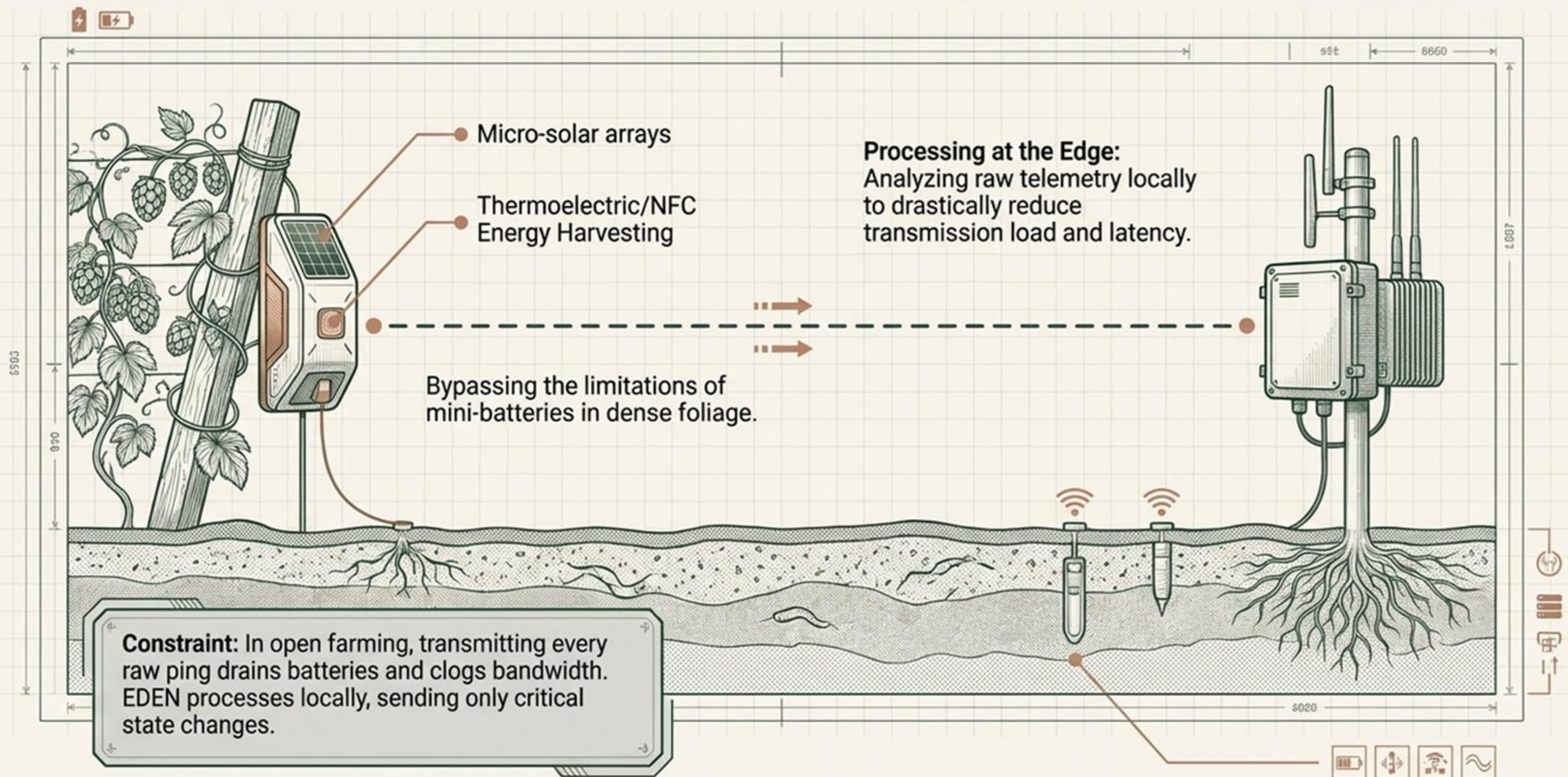
Project Éden Hops maps the physical farm as a dynamic grid of individual, high-resolution micro-climates.

The Edge Transition: Taming Heterogeneity

Biology is noisy. Different sensors emit data at incompatible intervals, scales, and formats. Edge Computing acts as a universal translator, fusing hyperlocal soil data with macroscopic imagery. The result is a purified stream.

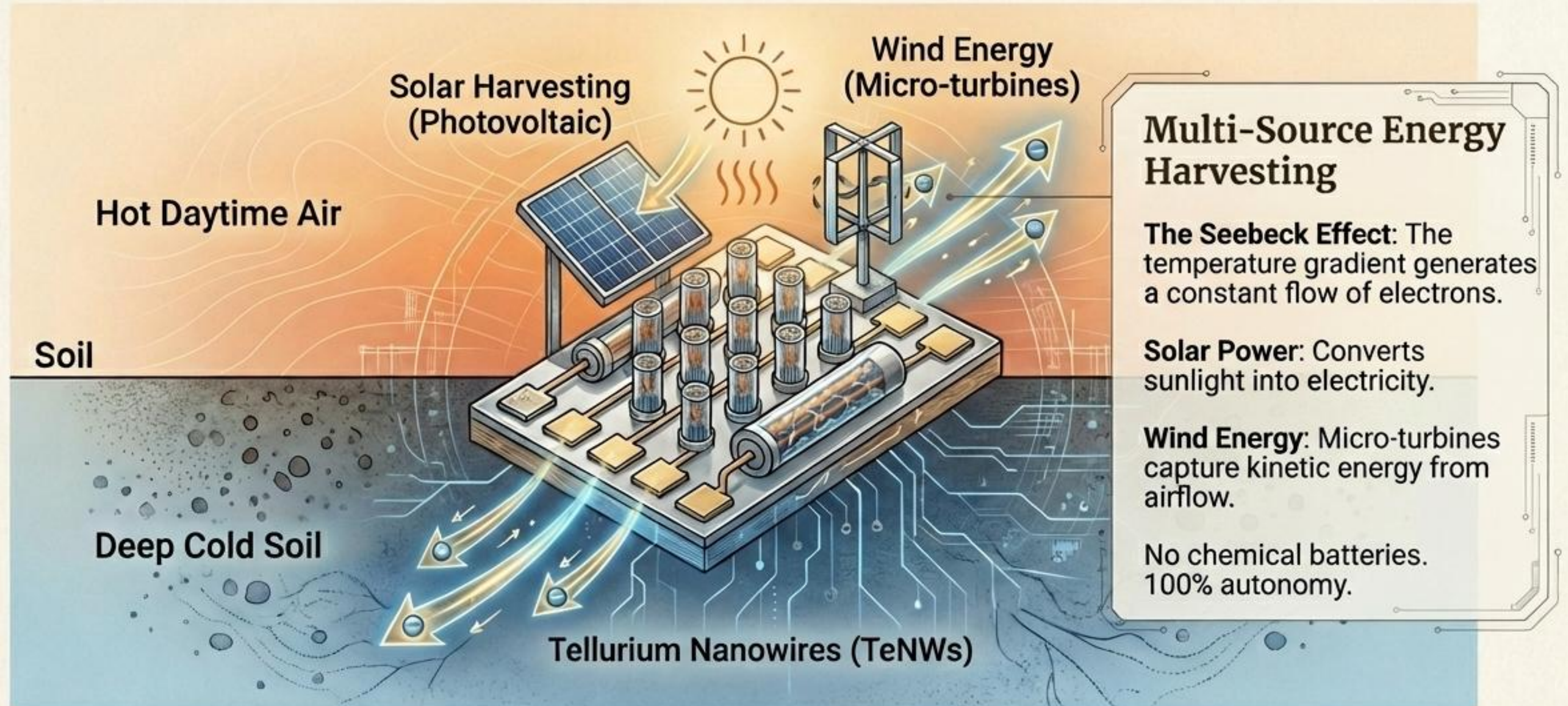


Solving the Open Field Energy Constraint

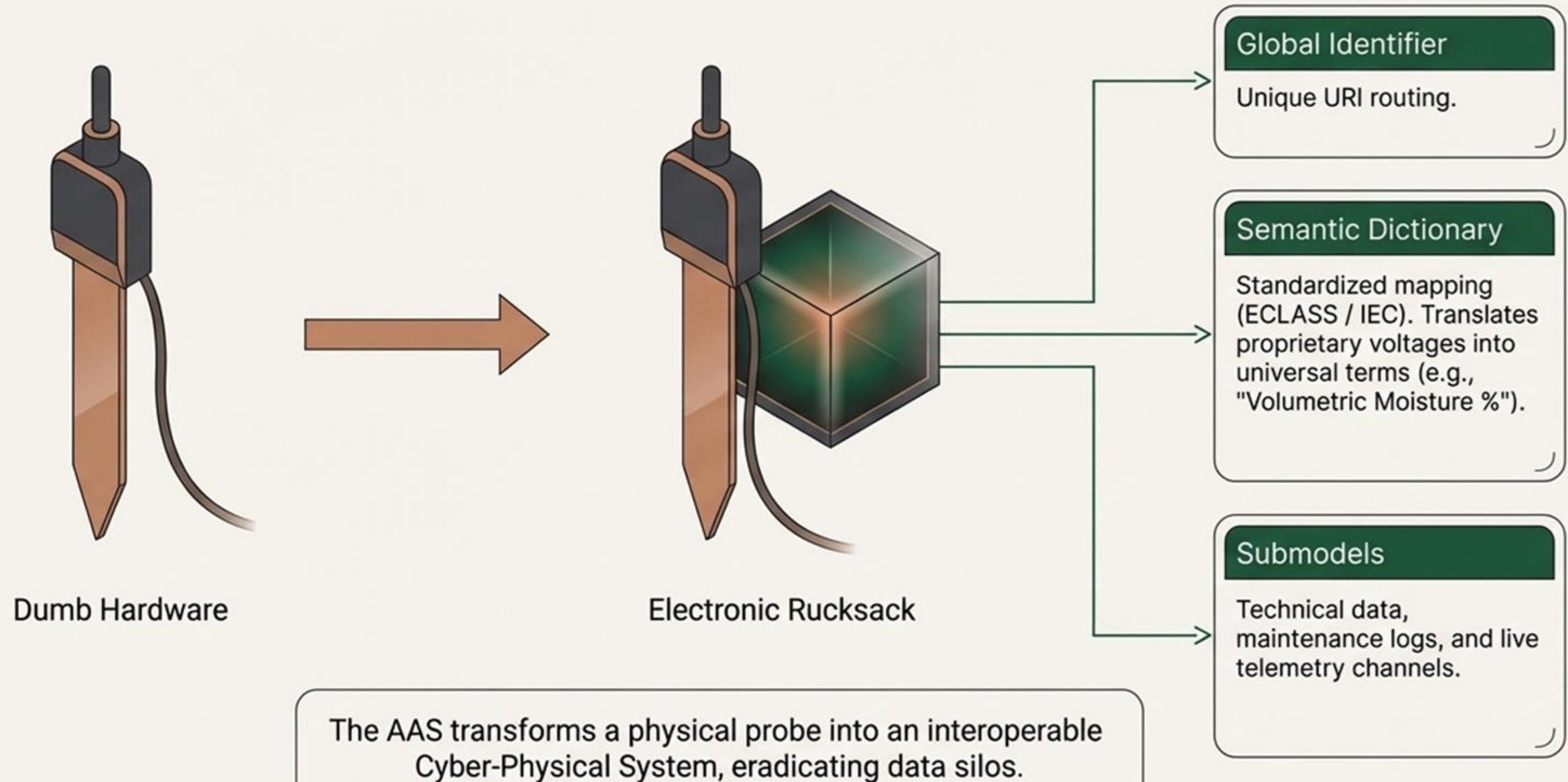


Infinite Autonomy in the Field

Wireless Sensor Networks (WSN) operate in vast areas where battery replacement is impossible. The solution is self-sustaining multi-source energy harvesting, including thermoelectricity, solar, and wind, enabling data nodes to operate uninterrupted.

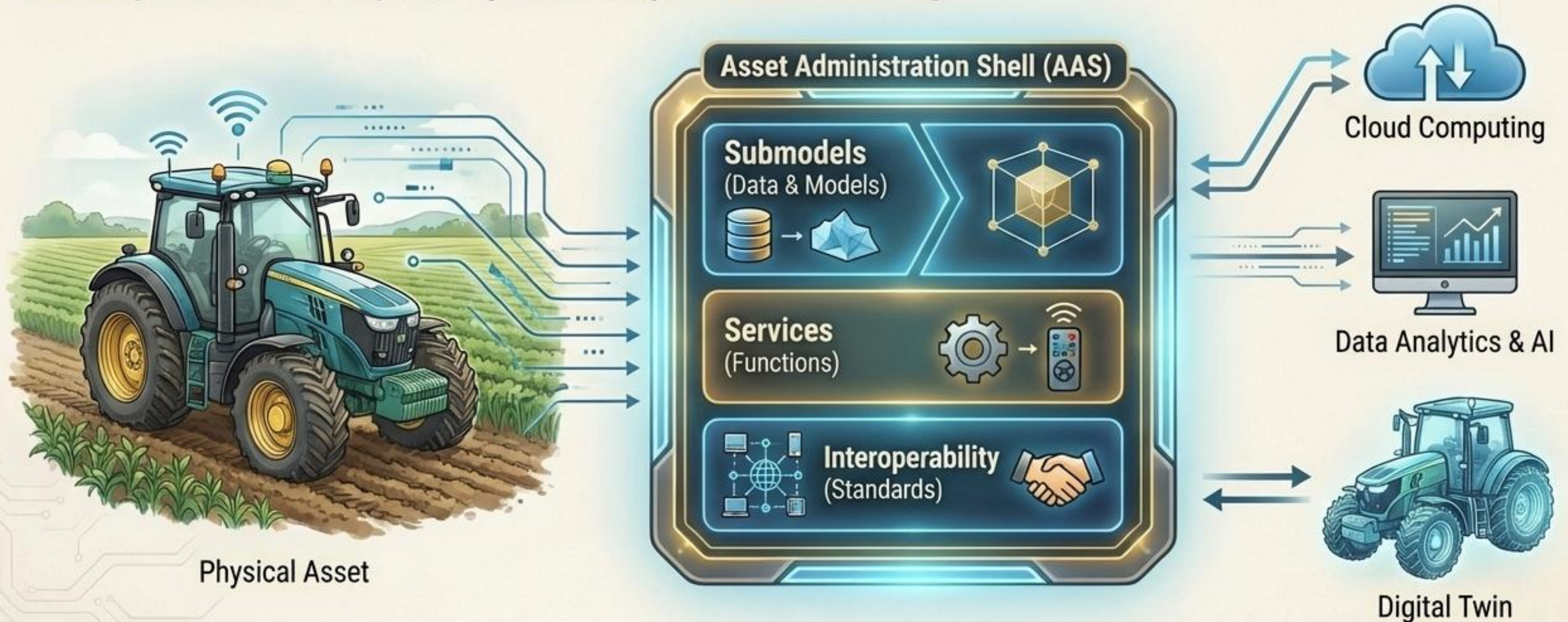


Step 2: Equipping the Asset Administration Shell (AAS)

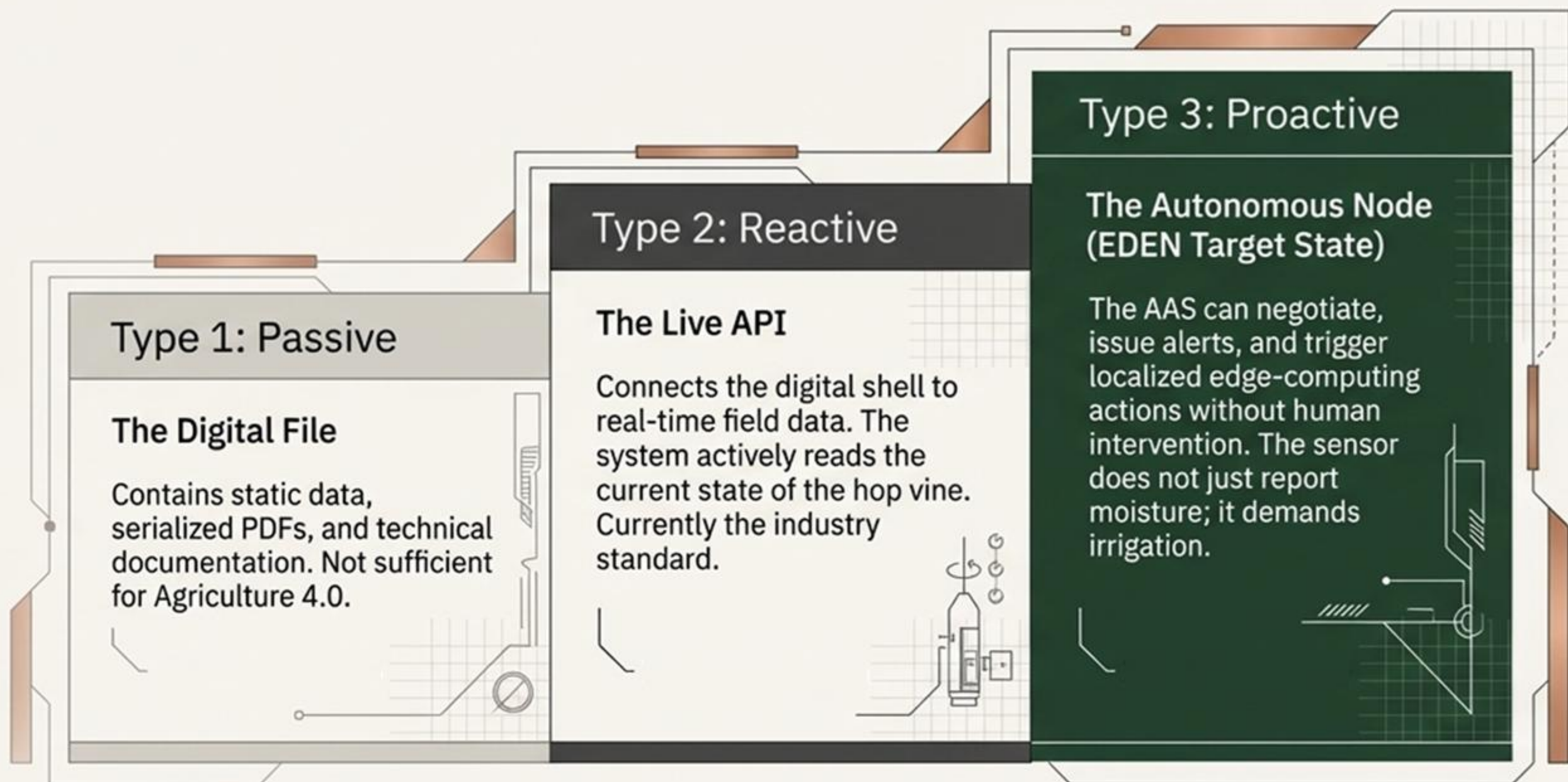


Agro in Industry 4.0: The Role of the Asset Administration Shell (AAS)

Industry 4.0 in agriculture requires a standardized digital representation. The Asset Administration Shell (AAS) acts as a digital twin, a structured container for all data, models, and services related to a physical asset, enabling seamless interoperability and intelligent decision-making.



The AAS Maturity Model: Toward Autonomous Integration



Step 3: The AI Processing Engine

Standardized AAS
Data Streams

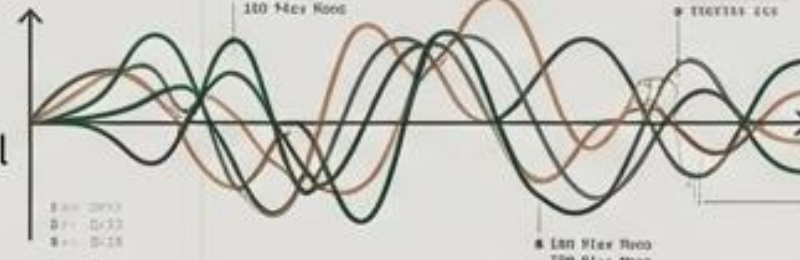
Weather
forecasts

Soil pH

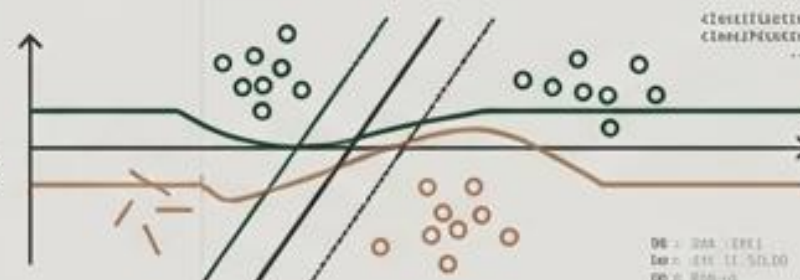
Historical
Hop Yields

Scikit-Learn Predictive Modeling

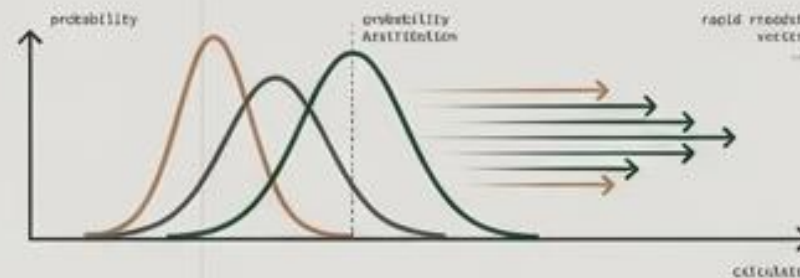
Random Forest:
Handling multivariable,
non-linear environmental
inputs.



**Support Vector
Machines (SVM):**
Classifying distinct plant
stress states.



Naïve Bayes: Rapid,
probabilistic weather-
impact calculations.



Actionable
Intelligence

Predicted
Biomass Yield

Optimal Irrigation
Timing

Pest
Vulnerability
Index

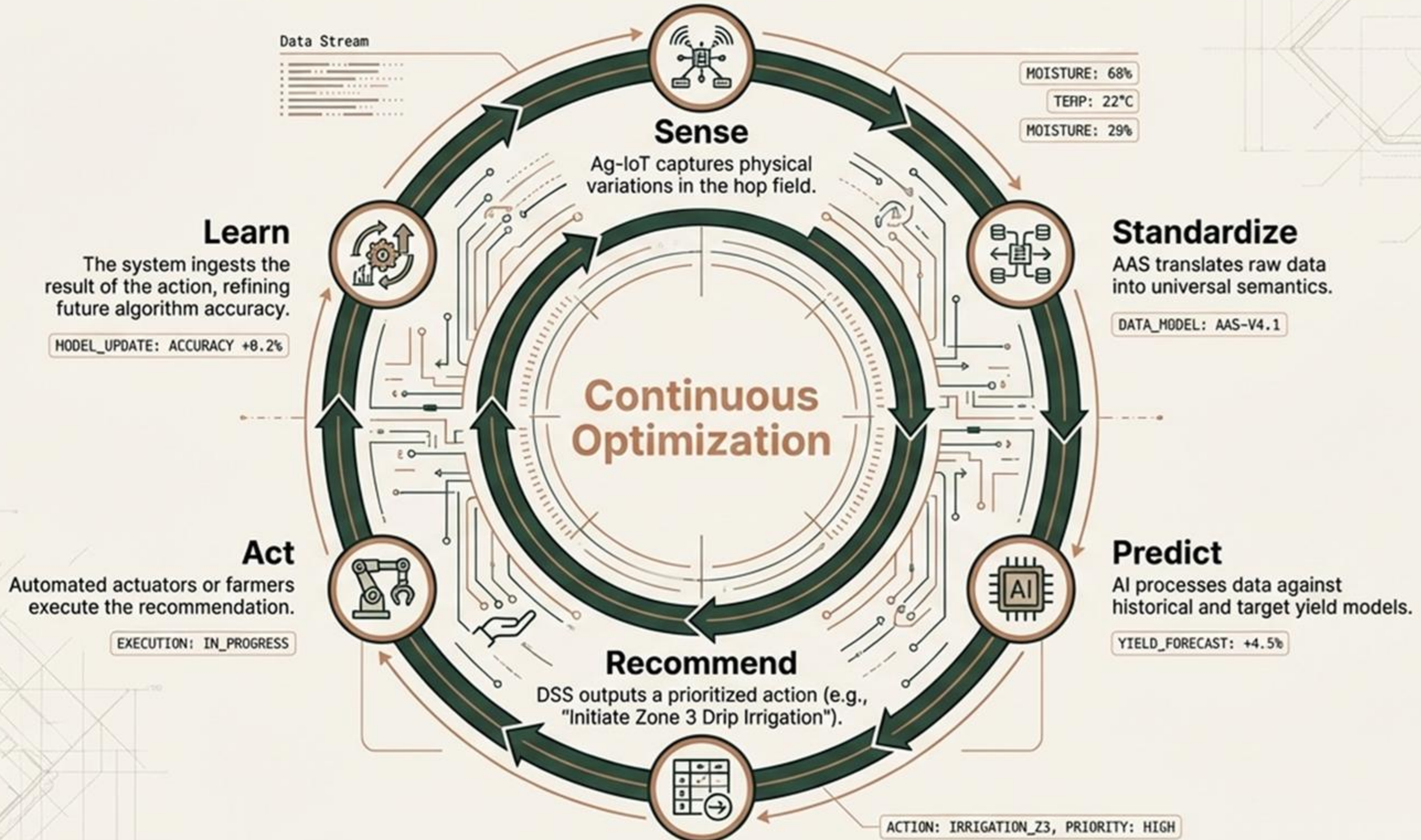
Executing What-If Scenarios in the DSS

Safe Experimentation

The DSS allows operators to inject simulated weather anomalies or adjust irrigation schedules, calculating the precise biological and economic impact before a drop of real water is spent.

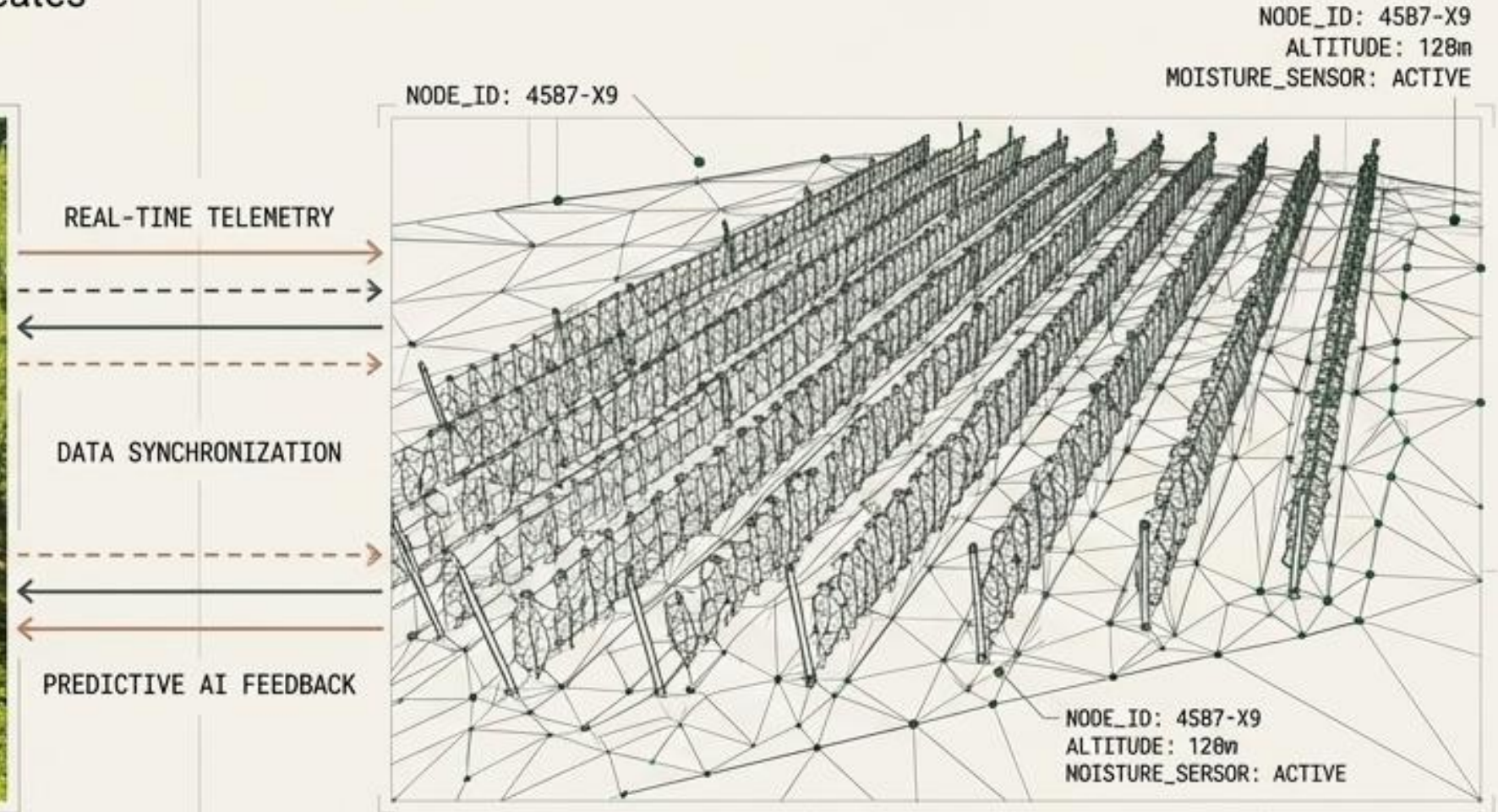


The Decision Support System (DSS) Feedback Loop



Synthesis: Awakening the Digital Twin

From Shadow to Model: By synchronizing continuous AAS telemetry with predictive AI, EDEN transcends simple monitoring. It creates a living, breathing simulated replica of the farm.



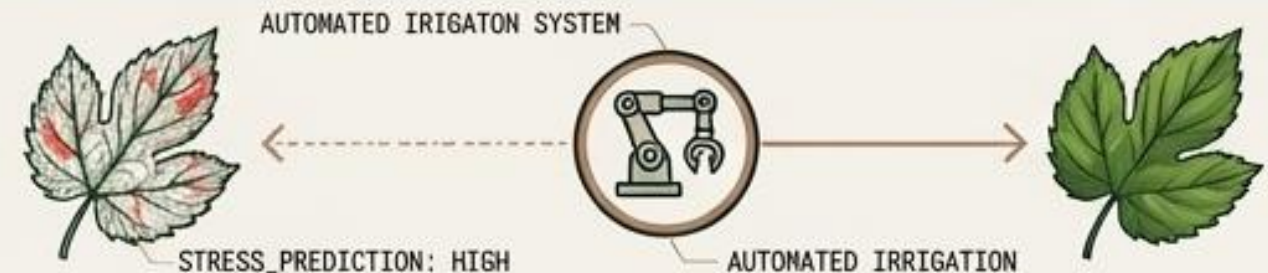
REAL-TIME TELEMETRY

DATA SYNCHRONIZATION

PREDICTIVE AI FEEDBACK

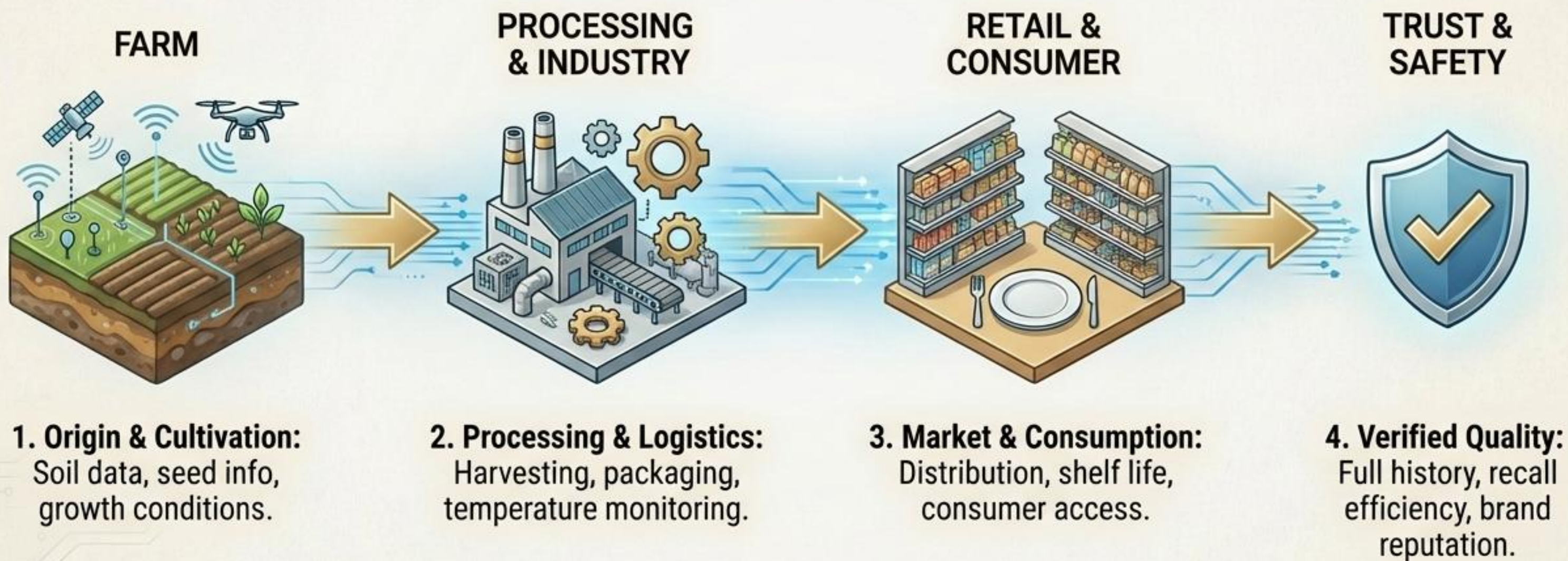
The Value Proposition:

We no longer manage the physical dirt. We manage the Digital Twin, observing how the virtual farm reacts to stress before a single physical leaf turns brown.



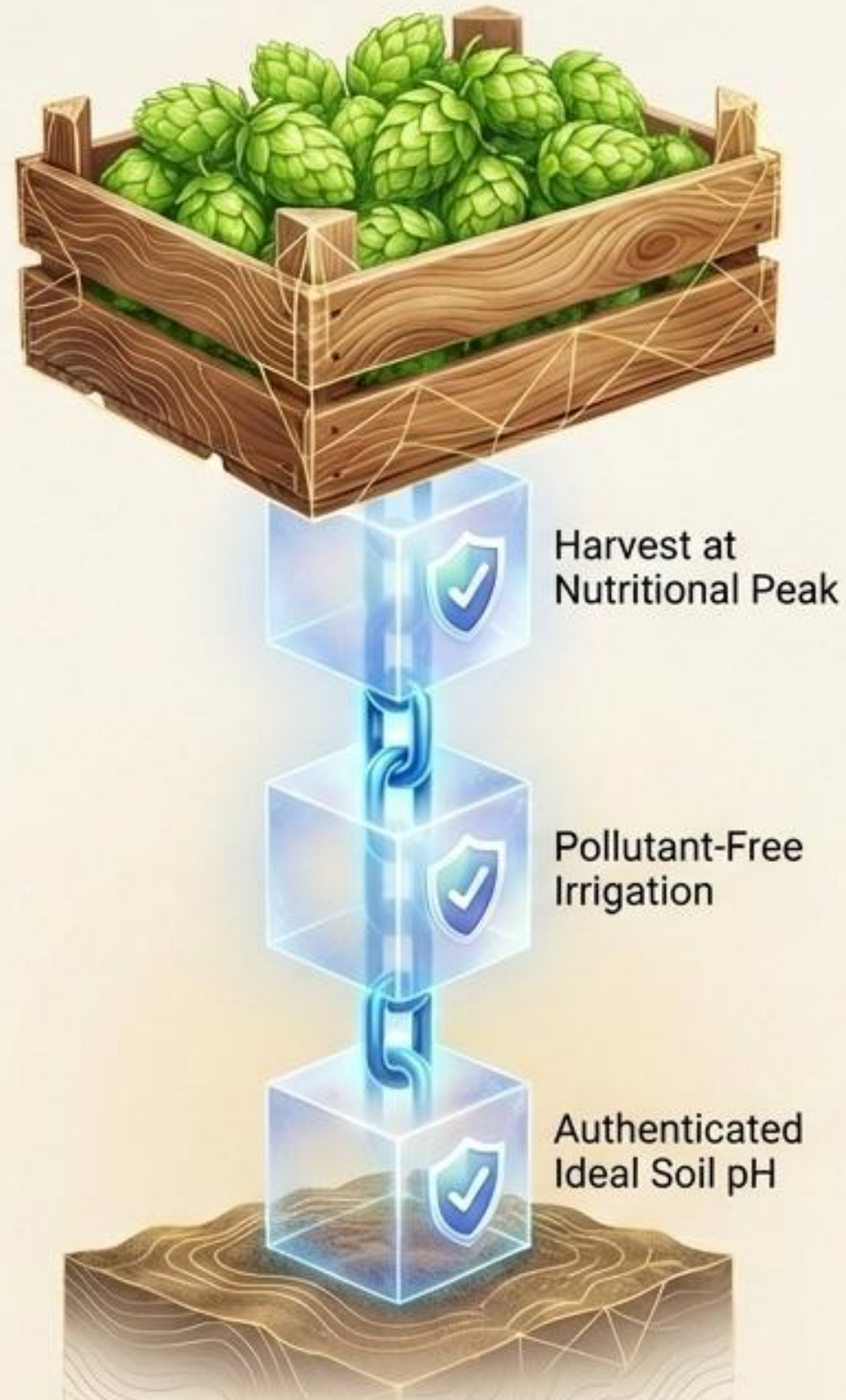
Farm-to-Table Traceability: A Transparent Supply Chain

Traceability is crucial for quality, safety, and trust. By capturing and linking data at every stage, from the soil to the consumer, we create an immutable record of the product's journey.



Blockchain and the Blocks of Trust

By recording sensor data directly onto a distributed and immutable ledger, we eliminate manipulation. The physical data collected from the seed mathematically guarantees the food's premium value on the supermarket shelf.



EDEN: The Blueprint for Scalable Agriculture 4.0



RESOURCE_EFFICIENCY: 99.9%

IBM Plex Mono
WASTE_ZERO_TARGET: ACTIVE

Ecological: Zero-Waste Resource Mapping

Hyper-precise DSS recommendations eliminate water waste and fertilizer runoff, protecting ground-water integrity.

RESOURCE_EFFICIENCY: 99.9%
WASTE_ZERO_TARGET: ACTIVE



YIELD_VARIANCE: <2%



SUPPLY_CHAIN_SECURE

Economic: Yield Certainty

Transforming volatile open-field hop farming into a predictable, industrialized production line, securing supply chains for local markets.



YIELD_VARIANCE: <2%
SUPPLY_CHAIN_SECURE: CONFIRMED



AAS_COMPATIBILITY: UNIVERSAL

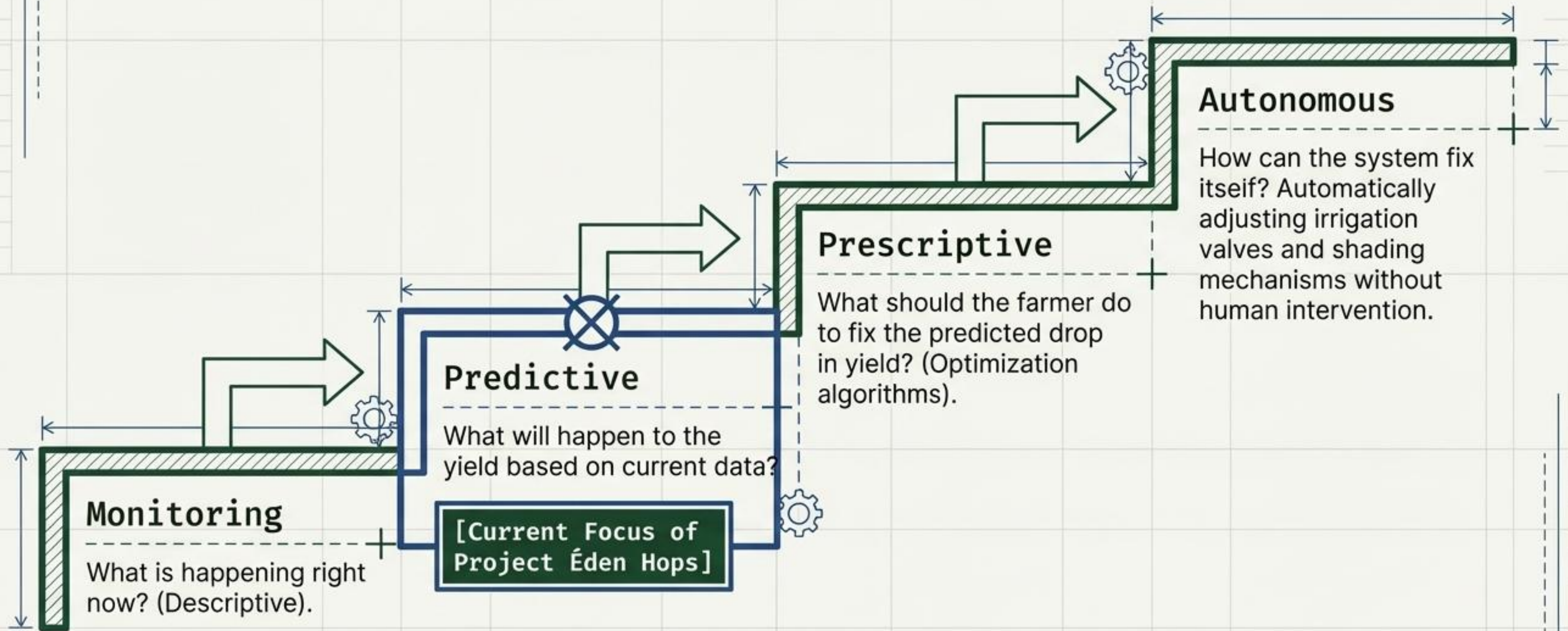
IBM Plex Mono
SCALE_FACTOR: GLOBAL_NETWORK

Technological: Infinite Interoperability

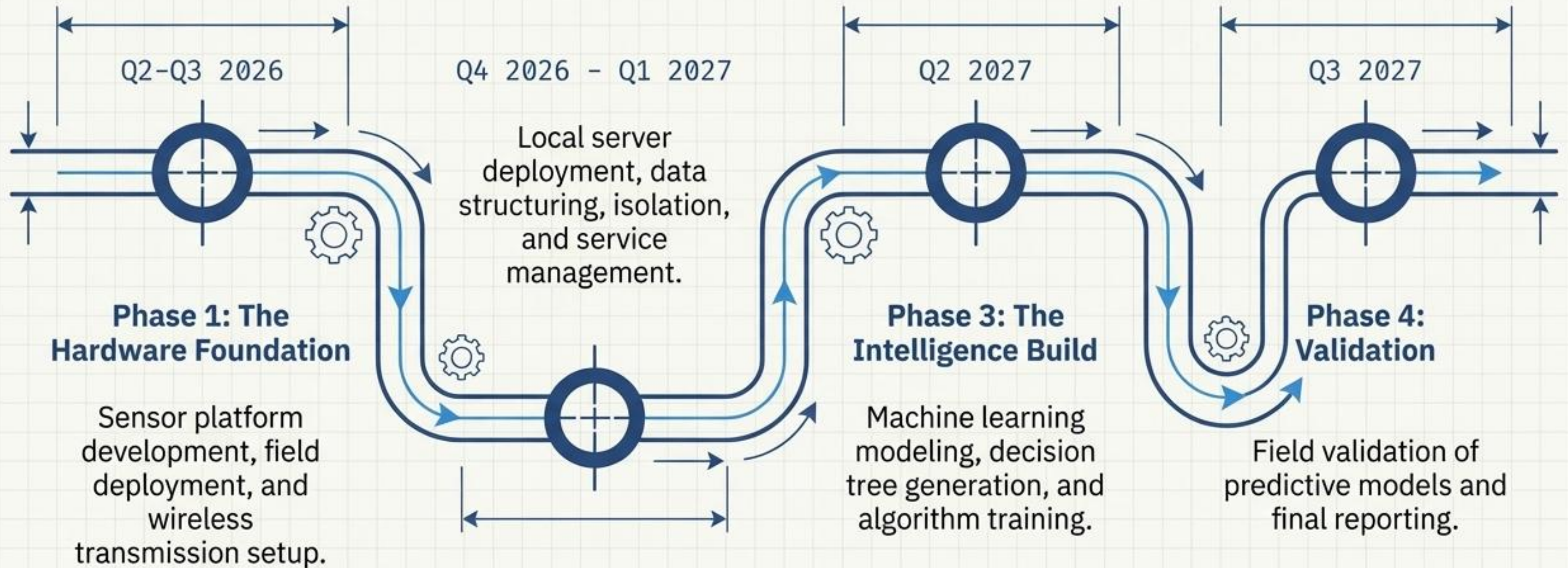
Built on open AAS standards, the EDEN architecture is universally adaptable, ready to scale from a single hop vine to global agricultural networks.

AAS_COMPATIBILITY: UNIVERSAL
SCALE_FACTOR: GLOBAL_NETWORK

The Digital Twin Maturity Spectrum



Execution Timeline (2026 – 2027)





The Future is Cultivated

- ☒ Merges Industry 4.0 standards (AAS) with practical agronomy.
- ☒ Transforms reactive, unpredictable farming into a predictive, data-driven science.
- ☒ Empowers local economies to reduce dependency on imported inputs.

Agriculture's next revolution doesn't just happen in the soil; it happens in the algorithm.